

- Promotional methods of online games keep changing.
- Integration of distinct scripts and plugins leads to complexity and increased development time and loading time as well, so we need a powerful platform with built-in features.

Proprietary vs HTML5

Usually, when we talk about 'proprietary vs HTML5', we're comparing developing games using only Flash with using HTML5—though many other proprietary tools exist. I won't denigrate Flash games in order to elevate HTML5 because Flash has been the key platform that changed the trend of online games. Most of the games we have got addicted to online, have been developed by using Flash. In fact, the term 'Flash games' has almost become synonymous with browser-based games, in general. For the end user, playing a Flash game requires installing a third party plugin to the browser. And on the development front, there is no source code provided for Flash games since its developer does not support open source. That's why HTML5 lovers say that 'It's not Flash, its Open.' Yes, this means that you never need to worry about whether Adobe is going to make crazy changes to the code sometime or the other. Every stage of HTML5 development is open for any developer to access the code.

Flash vs HTML5

Given the current sector trends and the several advantages HTML5 has over Flash, the future for HTML5 games looks bright.

Since we can't imagine the revolutionary development of online games without Flash, it would be better to evaluate HTML5 on its own merits rather than considering it as a substitute to Flash, since every tool has its own road map.

Hopefully, soon designers will no longer be forced to choose from a select few proprietary technologies that are not only expensive, but also obligate their end users to download, install, and regularly update the respective plugins. HTML5 is basically powered by JavaScript and CSS3 scripts, which are native to the browser. They work together and their collective power is increasing now, empowering the browsers!

Due to the rapid changes in online game building and usage, developers need the freedom to develop everything in one place and execute directly without installing any tools, and be able to play on any kind of platform as well.

The new element should be able to include maximum functionalities within, without depending on any external options to integrate action with the help of Flash, Silverlight, Flex, etc. The new standard - HTML5 - has many features that will replace some of the work Flash has performed over the years.

Major technical aspects of HTML5

With HTML5, the browser has become a full-fledged gaming platform today.

HTML5 represents a single standard that can be applied across multiple platforms, i.e., all computers and mobile devices.

Browsers are a lot smarter now and they are only going to continue getting smarter. Rather than relying upon plugins to display interactive and graphically intense pages, modern browsers can do it all by themselves.

In their most basic form, HTML5 games consist of JavaScript with the new HTML5 Canvas element, as well as cleaner, neater code. We can remove most *div* tags and replace them with semantic HTML5 elements like *aside*, *article*, etc.

Here are a few built-in features of HTML5 that are useful for developing games:

- Core Platform: JS (works with Canvas)
- Highly interactive and easy to use
- Less dependence on third party plugins
- Greater consistency
- Improved instant accessibility
- Highly portable (for devices)
- Great level of platform compatibility (same code base)
- Superb support of CSS3
- High performance graphics in 2D and 3D
- Reduced page load times and bandwidth usage
- Multi-touch support
- GPU powered HTML5 Canvas supports WebGL and SVG
- Offline asset storage and rich audio APIs
- File/hardware access (deeper integration with the OS)
- Client-side database enabled
- Reliable storage options in true SQL db instead of cookies
- Is the foundation of the upcoming mobile gaming era
- WebSocket-based networking for multiplayer capability
- Realtime communication enabled
- Geolocation support
- Great for social games
- It is not dependent on app stores

For getting better insight on production planning and developing a browser based game with HTML5, keep reading the upcoming articles.

To better understand the complete process of game development in HTML5, you can attend my upcoming workshop at OSI Days this year in November at Bengaluru, based on the topic: 'Developing a platform independent game in HTML5 for the Web and the mobile'. The detailed agenda of the workshop, essential background knowledge required to develop the game 'Feed the Bubble', along with the references for this article are supplied at www.mtc.cgmanifest.com 

By: Pradeep ISN

The author is the founder of the Production and Research Divisions of the CGManifest Group and has been its director since 2006. He is also the director of the international award winning School of Computer Graphics since 2008. With nearly 10 years' experience in Web technologies, digital film compositing, as well as administration of a production consulting unit and research oriented academic environments, he is now exploring innovative possibilities in HTML5-based gaming and interactive trends in education.